UNIT-III

Corrosion Science and Management

Abstract

Metals and alloys are essential to various engineering disciplines due to their remarkable properties like conductivity, malleability, ductility, and luster. Steel is vital for civil engineers, aluminum and its alloys for automobile engineers, and copper for electrical and electronics engineers. However, metals like iron, aluminum, and copper are naturally prone to corrosion, which necessitates protection to ensure their durability. Just as a doctor must understand a disease before treating it, a corrosion engineer must comprehend the mechanisms of corrosion to effectively address it. Often compared to cancer, corrosion has numerous triggers. This chapter explores the factors that lead to corrosion, examines key control techniques, and unpacks the chemistry underlying these processes.

Blow-up Syllabus

Sl No	Topic to be taught	Duration
1	Introduction to corrosion, Electrochemical theory of corrosion	1 hour
2	Types of corrosion: differential aeration (pitting and water line), differential metal and stress corrosion.	1 hour
3	Factor affecting rate of corrosion-electrode potential (nature of the metal), Nature of corrosion product, numerical problems on PB ratio.	1 hour
4	Factor affecting rate of corrosion -continue-pH of the medium, temperature, relative anodic and cathodic area (Area effect).	1 hour
5	Corrosion control: Metal coating-galvanization and tinning, surface conversion coating - anodizing and phosphating.	1 hour
6	Cathodic protection - sacrificial anode method. Corrosion testing by weight loss method. Corrosion penetration rate (CPR)-numerical problems	1 hour
7	Metal finishing: Introduction to Electroplating, electroplating of chromium.	1 hour
8	Introduction to electroless plating, electroless plating of copper.	1 hour
9	Case studies on corrosion failure.	Self learning

After going through this chapter, student *will*

(i) be able to understand and explain many of the corrosion situations in day to day life.

(ii) be able to predict corrosion in a metallic structure based on chemistry principles.

(iii) be able to suggest a proper method for corrosion control.

CORROSION

Introduction: Have a read at the following situations

A highly reactive metal like Lithium kept in vacuum, does not corrode.

(i) A reactive metal line sodium kept in kerosene also does not corrode.

(ii) When these metals are kept in air they vigorously react (corrode) to form their oxides.

The chemical reaction involved in 3 in presence of oxygen can be represented as below.

$4Na + O_2 \longrightarrow 2Na_2O$ ----------(1)

$2Li + O_2 \longrightarrow Li_2O$ ----------(2)

In above reaction 1, there is no environment whereas in reaction 2 there is kerosene environment. But corrosion is found to occur in case 3. What does this indicate? It makes it clear that a **suitable environment** is required for a metal to react and corrode.

Observe what has happened to the metals? They are **oxidized**. The products have entirely different properties than metals. So metallic properties are lost, and metal is destroyed. Then the metal is said to have corroded. The attack on the metal has been found to take place either electrochemically or by direct chemical attack (will be discussed later). Taking all these points into considerations the definition of corrosion can be written.

Corrosion is the destruction and consequent loss of a metal or an alloy due to chemical or electrochemical attack on its surface by its environment.

Don't forget that during corrosion **metal is oxidized** or whenever metal is oxidized it is said to have been corroded. A metal may give different types of products depending upon the environment. For example, Fe may give $Fe(NO_3)_3$ in nitric acid environment and $Fe_2O_3.H_2O$ in moist atmosphere. In both the cases it is corroded, as it is oxidized. In many of the batteries, at anode, metal gets oxidized.

Consider the rusting of iron in two different environments.

Case (i): Iron exposed to dry HCl or red-hot iron exposed to air

In these cases, metal is attached directly by the reactive environment. A rapid reaction occurs, and corrosion product can be observed almost immediately. The chemical reactions involved may be written as

$Fe + 2HCl \longrightarrow FeCl_2 + H_2$; $Fe(\text{red hot}) + O_2 \longrightarrow Fe_2O_3 / Fe_3O_4$

(Fe_2O_3 is commonly referred to as scale)

Case (ii): Iron exposed to normal atmosphere:

Consider a fresh metal piece susceptible for corrosion that has been kept exposed to air. It takes some time to observe any corrosion product on its surface. Here a slow reaction occurs leading to the formation of corrosion product. The moisture in air also plays an important role and oxygen

in the air attacks the metal with the aid of moisture. The reaction can be written as

$2Fe + 3H_2O + 1/2\ O_2 \longrightarrow 2Fe_2O_3\ 3H_2O\ [= Fe(OH)_3]$

Case 1 is considered as **dry corrosion** and the second one as **wet or electrochemical corrosion**

As mentioned earlier, corrosion involves oxidation of the metal ($M \longrightarrow M^{n+} + ne^-$). Metal cannot simply generate electrons unless there is another chemical available (in the environment) to absorb them (undergo reduction). Oxidation and reduction always occur simultaneously. A rich man cannot give money unless there is a poor to accept it would fit very well as an analogy. If the chemical is available at the place where the electrons are generated, they would be immediately absorbed by the chemical and the corrosion product is formed at that spot (Dry corrosion). If not, electrons migrate (electricity flow) to the point where, chemical is available and get absorbed (Electrochemical corrosion). The corrosion product may be formed at a place different from the point of oxidation.

The rate of the chemical reactions in the above cases indicates that their mechanisms are different. One can differentiate the two as below.

(1) Dry Corrosion or direct chemical attack:

This type of corrosion that occurs in the absence of any liquid medium (Hence the name dry corrosion).

- It is relatively a rapid process.
- Corrosion product is formed at the place of oxidation.

Attack of acidic gases like SO_2, NO_2 etc. on a metallic surface are the other examples for the same. When iron is in red hot condition one cannot expect a liquid medium to exist at the surface and therefore corrosion in this condition is also an example of dry corrosion

(2) Wet corrosion or electrochemical corrosion:

This type of corrosion occurs in presence of a liquid medium and most of the times it is moisture (water)

- It is relatively a slow process.
- Corrosion product is usually formed at a place different from the point of oxidation.

Corrosion of any metal kept exposed to air becomes an example for this type of corrosion. As this is very common, let us try to understand theory and the mechanism of this process.

Wet or electrochemical theory very well explains the Corrosion of most of the metals exposed to open air. Before discussing the theory there is a need to understand how an electrochemical cell works, as they are comparable. A cell consists of an anode and a cathode in contact with one another and both also in contact with a conducting solution. In this situation there will be

spontaneous oxidation at anode and reduction at cathode. The electrons liberated at anode migrate to cathode through the external circuit accounting for electric current. Such cells are generally called as electrochemical cells or galvanic cells.

Most of the metals or metallic structures are exposed to open atmosphere. One can find oxygen as well as moisture in atmosphere. According to this theory, corrosion in such environments is an electrochemical process involving migration of electrons and ions. Moisture in the environment can settle on the metal surface. Under this condition, according to this theory, corrosion occurs due to

Formation of tiny electrochemical or galvanic cells-Formation of anodic and cathodic areas on metallic surface and both these areas in contact with a conducting liquid medium (Most of the times conducting medium is water, refer to Figure 3.1). How anodic and cathodic areas are formed would be discussed later. The metal acts as electronic conductor and the liquid medium (mostly water) helps in the conduction of ions. There will be spontaneous oxidation and reduction reactions leading to the formation of corrosion product.

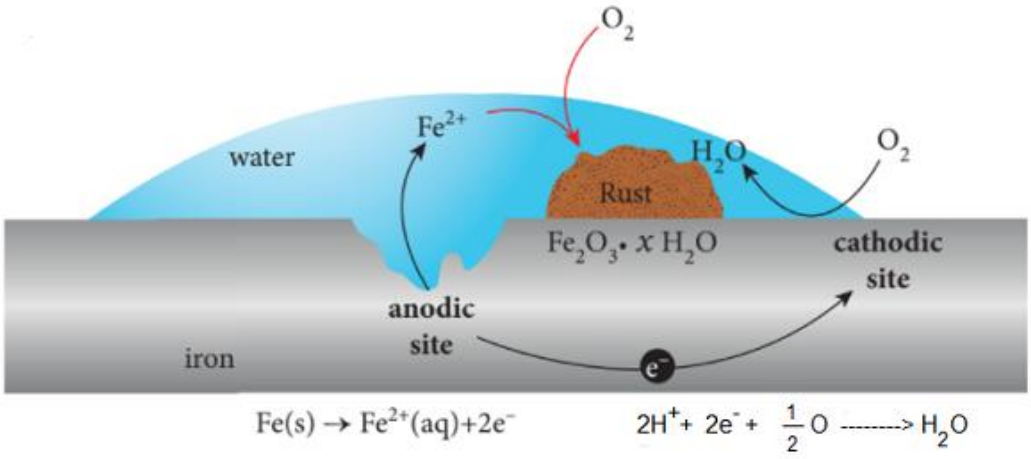


Fig 3.1: Formation of Electrochemical cell on a metallic surface

Let us look at the possible reactions at anodic area and cathodic area taking Iron as an example.

At Anodic area- Metal gets oxidized

$$M \longrightarrow M^{n+} + ne^-$$

$$Fe \longrightarrow Fe^{2+} + 2e^-$$

Metal ions so produced enter the liquid medium at the surface. The electrons produced migrate to the cathodic area through the metal itself as it is an electronic conductor. They are absorbed at cathodic areas (through reduction reaction) by the species present in the environment.

At Cathodic Area- The environment (the liquid medium) on the metal surface can be acidic, alkaline or neutral. Further it may or may not contain dissolved oxygen. Depending upon the

nature of the environment, electrons are absorbed by different mechanisms. Let us consider all possible conditions of the medium and write the chemical mechanisms for electron absorption.

a. Hydrogen evolution mechanism at cathode (In absence of oxygen)

(i) When the medium is acidic (H^+ concentration in more) and there is no dissolved O_2 in it, then.

$$2H^+ + 2e^- \longrightarrow H_2\uparrow$$

(ii) When the medium is alkaline or neutral and there is no dissolved O_2 in it, then.

$$2H_2O + 2e^- \longrightarrow 2OH^- + H_2\uparrow$$

b. Oxygen absorption mechanism at cathode (in presence of oxygen)

(i) When the medium is acidic (H^+ concentration in more) and there is dissolved O_2 in it, then.

$$4H^+ + O_2 + 4e^- \longrightarrow 2H_2O$$

(ii) When the medium is alkaline or neutral and there is dissolved O_2 in it, then.

$$2H_2O + O_2 + 4e^- \longrightarrow 4OH^- \text{ (this is the most probable reaction)}$$

This is how the theory explains the mechanism of oxidation and reduction reactions during corrosion.

Fe is the metal corroding in neutral or alkaline environments in presence of oxygen, the chemical reactions involved in rust formation can be represented as

$$Fe \longrightarrow Fe^{2+} + 2e^- \text{ (At anodic area)}$$

$$2H_2O + O_2 + 4e^- \longrightarrow 4OH^- \text{ (At cathodic area)}$$

$$Fe^{2+} + 2OH^- \longrightarrow Fe(OH)_2\downarrow \text{ (insoluble and therefore settles)}$$

$Fe(OH)_2$ is not stable . It gets further oxidized by oxygen.

$$4Fe(OH)_2 + O_2 + 2H_2O \longrightarrow 4Fe(OH)_3 \text{ or } Fe_2O_3.3H_2O\text{(Rust)}$$

Points to note:

1. Oxidation and reduction are two simultaneous processes.
2. If the environment does not contain any species that can absorb electrons (undergo reduction) then corrosion of the metal won't occur. That is the reason why a metal kept in vacuum or sodium kept in kerosene does not corrode.
3. Metal will not corrode in a cathodic area as it is not involved in any corrosion reaction whereas it gets eaten away only at anodic area.

When are anodic and cathodic areas formed on a metallic surface?

The understanding of electrochemical theory and the corrosion process is incomplete if one does not understand the condition under which anodic and cathodic areas are formed i.e when and how they are formed? Some of the conditions are listed below.

1. **When two or more metals are in contact with one another**
   It is called **galvanic corrosion**. As each of the metal has its own tendency to undergo oxidation, the metal with higher tendency to undergo oxidation (with lower reduction potential) becomes anodic and corrodes.
2. **Whenever a metal is exposed to different oxygen or air concentration**-it is called **differential aeration corrosion**. The less oxygenated part becomes anode and corrodes.
3. **Whenever some area of a metal is under stress**- it is called **stress corrosion**. The metallic part under stress becomes anode and corrodes.
4. **When the metal in impure**- the impurity may act as anode or cathode
5. **When different phases of a metal are at the surface**- A metal can exist in different phases and each of the phases will have different properties. Two such phases at the surface lead to formation of anodic and cathodic areas.

Types of corrosion:

There are a good number of reasons / properties associated with metal as well as the environment that decide corrosion and its rate. Specific names have been given to them and are being considered under **types of corrosion**. Most common types of corrosion are

Galvanic corrosion or Differential metal corrosion:

A Daniel cell ($Zn/Zn^{2+}//Cu^{2+}/Cu$) consists of two metals in contact with each other and in contact with a liquid conducting medium. It is also called galvanic cell. Zinc spontaneously undergoes oxidation (corrosion) and corrosion under such a situation is termed **galvanic corrosion**.

Corrosion observed when two dissimilar metals are in contact and also in contact with a conducting solution (mostly water) is called galvanic corrosion or differential metal corrosion.

The examples below (Fig.2) illustrate such situations that can exist in a designed metallic structure.

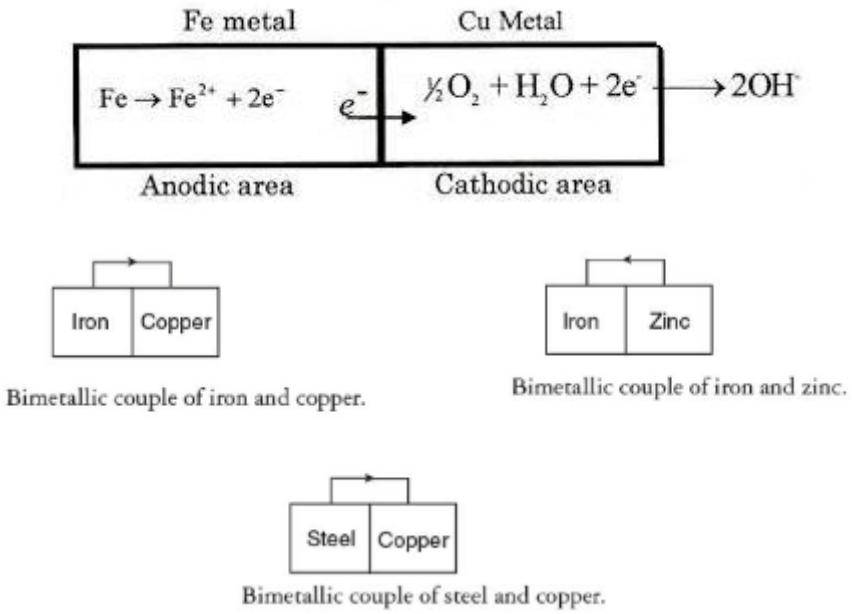


Fig 3.2. Some examples of differential metal corrosion

It is clearly known that metal corrodes at anodic area. When two or more metals are in contact one of the metals that acts as anode, corrodes. Which metal corrodes can be predicted with the help of the following.

a. The metal with higher tendency to get oxidized (with lower reduction potential) becomes anode and corrodes. With no other factor contributing, corrosion tendency can be guessed by its position in the electrochemical series (in which the elements are arranged in the increasing order of their standard reduction potentials). When two metals are in contact, the one which is upper in the series becomes anode and corrodes.

b. The other metal becomes a cathode. The reduction reaction at cathode depends upon the nature of the environment. Example: when Zn (standard reduction potential = -0.76V) is in contact with Cu (standard reduction potential = 0.34V), Zn becomes anode and corrodes. Similarly, when Cu is in contact with Ag (Standard reduction potential = 0.76V), Cu becomes anode and corrodes as Cu has a lesser reduction potential.

   In the fig .3, iron corrodes, and nothing happens to copper.

What decides the rate of galvanic corrosion?

The rate at which the metal that acts as anode corrodes depends upon

a) The difference in the electrode potentials of the two metals- larger the difference more is the rate of corrosion. For example: Zn corrodes faster when it is in contact with silver than in contact with Cu in a given environment.

b) Relative areas of anode and cathode- large cathodic area and small anodic area makes the anodic metal to corrode faster.

c) Environmental factors like pH, temperature, concentration of oxygen, oxidizing agents etc.

d) Passivation of the metal- Corrosion product formed usually settles on the metal surface as a layer. If the layer formed covers the metal completely and hence prevents the environment from coming in contact with the metal, the metal stops corroding. The metal is said to have become PASSIVE. Passivation is observed in case of metals like Al, Ti, Cr etc.

e) Polarization either at anode or at cathode or at both (discussed later under factors)

How to minimize galvanic corrosion?

If the use of more than one metal is inevitable in a structure galvanic corrosion is expected. In that case corrosion can be minimized by the following methods.

a. Proper selection of the metals- the metals should have least difference in their electrode potentials (very close in electrochemical series or galvanic series)

b. Taking care to see those small areas like screws, bolt-nut etc. in the structure become cathodes but not anodes. That means maintaining a small cathodic area compared to anodic area.

c. As far as possible by preventing the contact between the two metals using proper insulating materials like washers, sleeves etc.

Differential Aeration Corrosion:

An arrangement of the type $M/M^{+n}(c_1M)\ ||M^{+n}(c_2M)/M$ is called a concentration cell. It is the one in which a single metal is exposed to two different concentrations. The part exposed to lower concentration becomes anode and corrodes. The concentration difference is expected to be with respect to one of the species involved in electrode reaction and it is oxygen or air in an open environment.

Corrosion of a metal due to difference in air or oxygen concentration on its surface is called differential aeration corrosion.

The less aerated area (having less oxygen concentration) acts as anode and corrodes. The more aerated area acts as cathode. One can have doubt why a more aerated area acts as cathode? The fact is that oxygen helps in the absorption of electrons (reduction) and it happens at cathode. One such reaction can be written as

$$2H_2O + O_2 + 4e^- \longrightarrow 4OH^-$$

So, wherever there is more oxygen, reduction gets favored and electrons are absorbed. As an example, let us understand how a water drop settled on a metallic surface (Fig. 3.3) can lead to differential aeration and hence to corrosion.

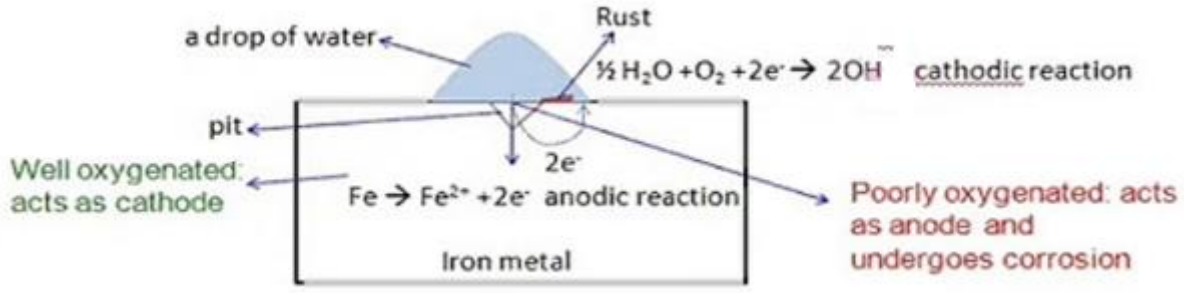


Fig 3.3. Differential aeration due to rust/water drop.

Specific names are attached to the differential aeration corrosion that occurs under different situations.

a. Waterline corrosion: It is a case of differential aeration corrosion. Consider a metallic part partially dipped in water. The surface of water (water line) will be continuously in contact with water and therefore is more aerated. The part of the metal just below the waterline (less aerated) acts as anode and corrodes. The part above the waterline acts as a cathode. Corrosion products are usually observed below the water level. Waterline corrosion explains the corrosion of a ship below the sea water level, corrosion of a water storage tank below the water level and Corrosion of a metal pillar or structure partially inside water (Fig. 3.4)

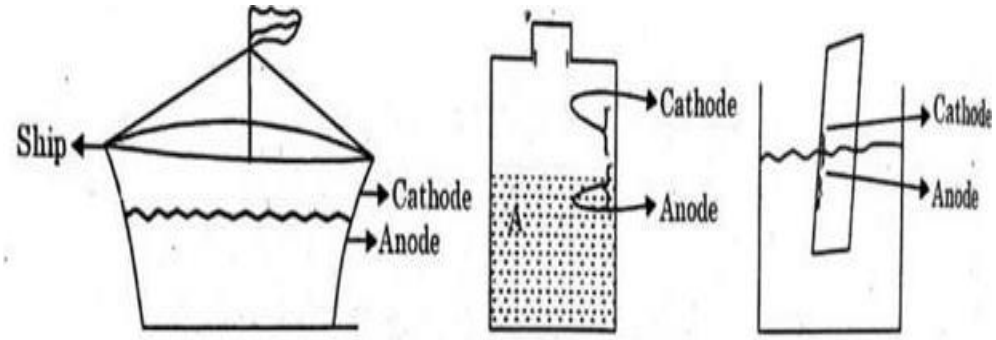


Fig 3.4. Some examples of waterline corrosion

b. Pitting corrosion: This is another case of differential aeration corrosion.

Consider a metal on which a small dust or rust or sand particle has settled (Fig. 3.5). The area below the particle automatically will be less aerated and becomes anode. The area around the particle acts as a cathode. Another important observation is that the area of anode is very small when compared to the cathodic area. Cathodic area will put up a higher demand for electrons that are to be supplied by a small anodic area. So, metal gets oxidized rapidly leading to the formation of a pit or hole. Hence the name pitting corrosion. Pitting

can be defined as **the localized and intense corrosion of a metal leading to the formation of a pit or pin hole around which the metal is relatively un attacked, called as pitting.** The product formed covers the anodic area making it more deficient in oxygen or air.

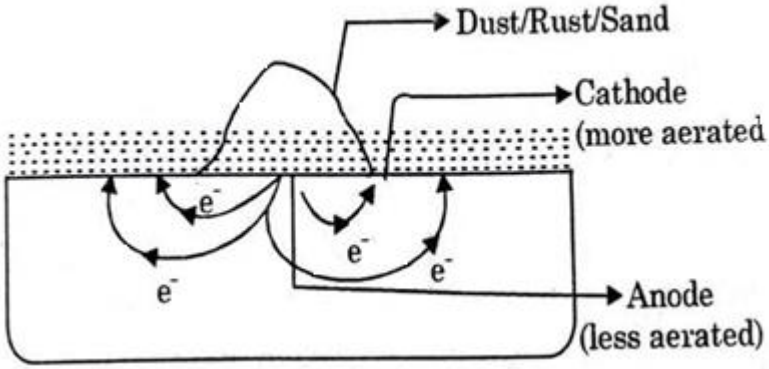


Fig. 3.5: Pitting corrosion

Any situation leading to a small anodic area and a large cathodic area leads to pitting. For example, Fe is protected from corrosion by coating it with another metal like Sn (tinning), chromium (chrome plating) etc. whenever there is a crack in the protective cathodic coating (Fig. 3.6), the cracks become anodes and the remaining surface of the metal would be cathode. This leads to pitting.

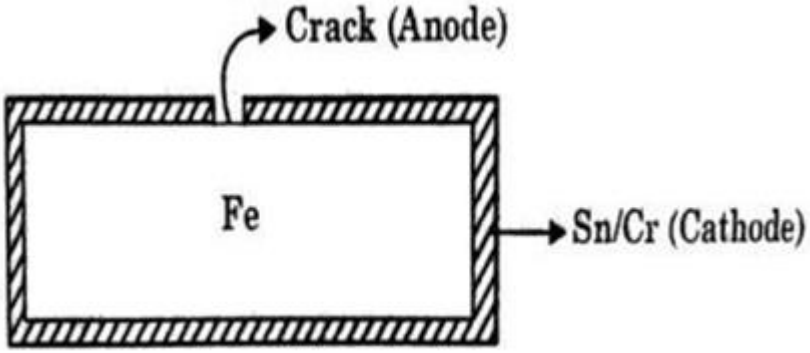


Fig 3.6: Cathodic coating

Stress Corrosion:

Metallic objects are fabricated using various mechanical operations like bending, pressing, drawing, hammering riveting etc (Fig. 3.7). During these operations the metal will experience stress. If the metallic part is not properly annealed, stress remains. The atoms and the grain boundaries at the stressed part exhibit greater reactivity (higher potential to react) compared to the unstressed part. They become anodes and start corroding. Stress corrosion is due to the combined effect of tensile stress and a specific environment.

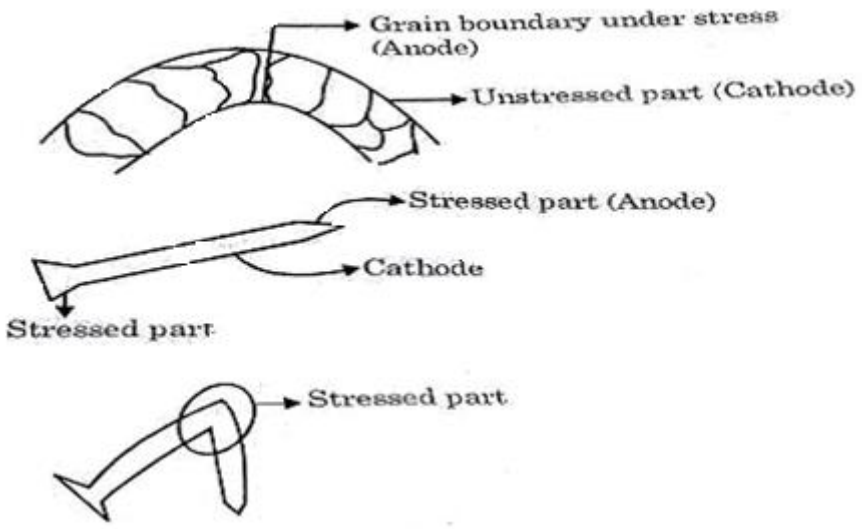


Figure 3.7: Stress corrosion

It is a common thing to observe that cracks appear at the stressed parts (for example if a nail is bent some 3-4 times at a point, a crack develops at the bent part) and it is more in the case of hard metals and alloys. In this situation the cracks act as anodes and corrosion starts. The cracks deepen into the metal, and it is also referred to as **Intergranular stress corrosion.** It is also observed that during stress cracking, the metal or alloy is mostly unattached over most of its surface except along the cracks.

Caustic embrittlement is a type of stress corrosion found to occur in mild steel boilers. To necessary conditions for this are

(i). Hairline cracks due to stress and (ii). Presence of caustic alkali (NaOH).

Let us try to understand it a little more.

Stressed part of a boiler will have fine cracks (Fig.3.8). Whenever water contains Na_2CO_3 it enters the cracks and undergoes hydrolysis to produce NaOH.

$$Na_2CO_3 + H_2O \longrightarrow 2NaOH + CO_2$$

This increases the concentration of NaOH in the cracks. As a result, a concentration cell of the type shown below is formed.

Fe (In the cracks) |Concentrated NaOH ||Dilute NaOH | Fe (unstressed part)

The metal dissolves at anode. The hair cracks therefore grow along the direction of the stress leading to the failure (sometimes explosion) of the boiler. The reactions in the cracks can be represented as

$$2NaOH + Fe \longrightarrow Na_2FeO_2 + H_2$$

$$3Na_2FeO_2 + 4H_2O \longrightarrow Fe_3O_4 + H_2 + 6NaOH$$

NaOH so generated attacks the metal in the cracks and the cracks develop.

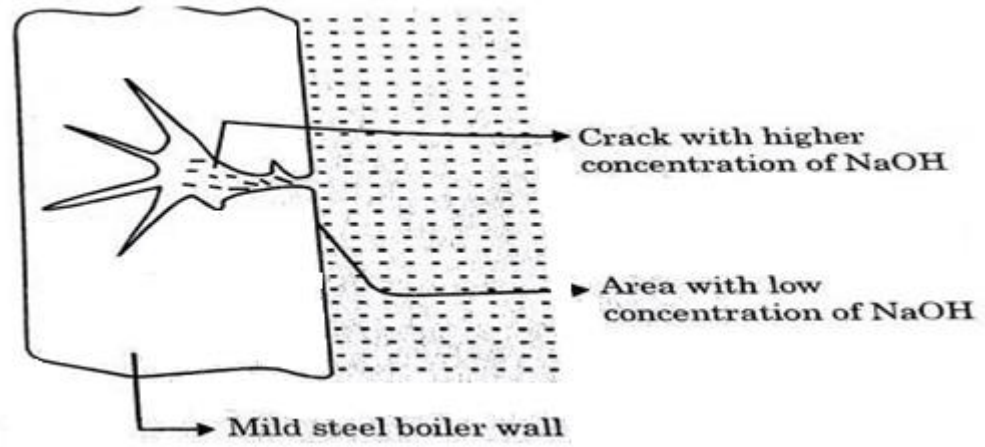


Fig. 3.8. Caustic embrittlement in steal boilers

Factors influencing the rate of corrosion

It is necessary to know the rate of corrosion of a metal as it decides the life of it. It is a fact that Iron corrodes faster than copper or brass. Also, Iron corrodes slowly in open atmosphere but corrodes faster in an acidic environment. Similarly, brass corrodes faster in ammonia environment than in open air. These examples indicate that the rate of corrosion depends upon the metal in use and its environment. Let us analyze some of the important factors that influence the rate of corrosion of a metal. Some are related to the metal itself (called primary factors) and some are related to the environment (called secondary factors)

1. **Nature of the metal (Electrode potential)**: some metals have inherent property to react (corrode) faster. For example, metals like Na, K, Mg corrode faster than Cu, Ni and Ag, in air. Electrode potential of the metal can be used to understand its corrosion tendency. Lower the reduction potential (means higher the oxidation potential), more is the tendency of the metal to get oxidized (corroded)

   Note:

(i) When two metals are in contact with one another, the difference in their potentials decides the rate of corrosion, in a given environment. For example, Fe corrodes faster when in contact with Ag than in contact with Cu (compare their reduction potentials)

(ii) Rate of Corrosion of some metals like Al, Ti, Cr cannot be predicted based on electrode potential alone as they show a tendency to undergo **PASSIVATION** (formation of a protective layer of oxide in air preventing further corrosion)

2. **Nature of Corrosion Product**: Corrosion product is the result of chemical reaction between the metal and the environment. A suitable environment is necessary for corrosion to occur. In an open atmosphere the corrosion product formed is usually an oxide or carbonate. This is because oxygen and carbon dioxide are the reactive chemicals in air. Sulphates and nitrates are also possible if air contains SO_3 and NO_2 respectively. But it is rare. The corrosion products formed on metal surfaces are usually insoluble in a normal environment and therefore settle on the surface forming a layer. The layer, even nanometer thick, if it forms a barrier (Figure 3.9) between the metal and the environment, then the corrosion stops or its rate decreases.

If the corrosion product is (i) insoluble (ii) Nonporous (iii). Adherent (iv) Non-volatile (v). Non-conducting and (vi)Stoichiometric, then it forms a perfect barrier layer between the metal and the environment. Such a layer passivates the metal from further corrosion. Al, Ti, Cr etc produce oxides that possess the above properties. Such metals exhibit passivity and stop corroding.

If the corrosion product is (i) soluble (ii). Porous (iii). Non adherent (iv) volatile (v) conducting and (vi) non-Stoichiometric then the barrier layer allows the environment to penetrate and reach the metal. So metal continues to corrode, may be at a slower rate. Since the oxide layers formed on metal like Fe, Cu etc. are porous and non-adherent these metals continue to corrode.

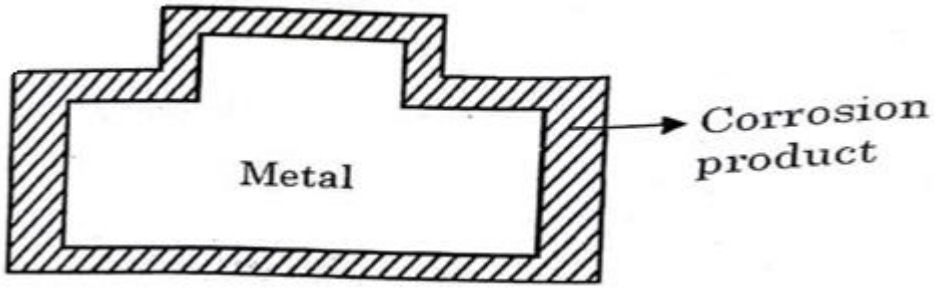


Fig 3.9: Nature of the corrosion product

Passivation is related to the volume occupied by the metal oxide compared to the volume of the metal. It is expressed in terms of specific volume ratio (volume occupied by unit weight).

$$\text{Specific volume ratio} = \frac{\text{Volume occupied by the metal oxide produced}}{\text{Volume occupied by the metal that has reacted}}$$

If this ratio is <1, as in the case of Mg, Ca, Na, Fe etc, there will be gaps or cracks in the protective barrier and the metal continues to corrode. If the ratio is >1 (which means that the specific volume

of the metal oxide is greater than that of the metal), it covers the surface well, forming a nonporous layer. This is the case with metals like W, Ti, Cr etc.

Pilling-Bedworth ratio (PBR)-Predicting passivation of a metal.

In most of the cases, when metal corrodes, metal oxide is the corrosion product. This is because oxygen is a reactive species in the environment. The corrosion product always settles on the surface of the metal. How well the settled product covers the metal surface will decide the rate of further corrosion. Pilling and Bedworth have studied this factor and proposed an expression using which the protection (Passivation) of a metal by the corrosion product can be predicted. When a metal corrodes corrosion product is formed. The volumes occupied by a definite amount of a metal and its corrosion product are different. This depends on the packing of atoms in the solid and in turn is related to the densities of the two. If the corrosion product is porous metal continues to corrode and if it is non-porous, metal stops corroding.

Pilling-Bedworth has found that the protective nature of the film (corrosion product) could be evaluated based on the ratio of the volume of the metal oxide to the volume of the metal that has produced the oxide. This is called a specific volume ratio and the statement is called the Pilling-Bedworth rule.

$$\text{Specific volume ratio} = \frac{\text{Volume occupied by the metal oxide produced}}{\text{Volume occupied by the metal that has reacted}}$$

According to PBR, if the specific volume ratio is **less than 1** (when the volume of the metal oxide is less than the volume of the metal) then the product can not completely cover the metal surface and therefore corrosion continues to occur. When PBR is greater than or equal to 1 and **less than 2**, there will be good protection (Passivation). When it is **more than 2**, a thick layer is formed that tends to peel off therefore good protection is not expected.

For alkali and alkaline earth metals the ratio is < 1 whereas for metals like Al, Ti and Cr the value is >1.

Calculation of specific volume ratio

Consider a general corrosion reaction,

$$M + 1/2O_2 \longrightarrow MO \text{ (Ex: } Mg + 1/2O_2 \longrightarrow MgO)$$

Here one atomic weight of a metal has produced one molecular weight (1mole) of corrosion product. if A and B are the atomic weight of M and molecular weight of MO and d_M and d_{MO} are the densities of the two respectively, then,

$$\text{Specific volume ratio} = \frac{B/d_{MO}}{A/d_M}$$

$$\frac{\text{Weight}}{\text{Density}} = \text{Volume}$$

If the corrosion reaction is

$2M + 3/2\ O_2 \longrightarrow M_2O_3$ (Ex: $2Al + 3/2O_2 \longrightarrow Al_2O_3$) then A has to be multiplied by 2. This is because two atomic weights of the metal are producing one molecular weight of the oxide. In general, if an atomic weight of a metal is involved then the weight of the metal to be taken as nA.

Problem: Predict the passivation tendency of a metal M that reacts with oxygen as per the following reaction

$M + O_2 \longrightarrow MO_2$, given atomic and molecular weight of M and MO_2 as 35 and 67 respectively. The densities of M and MO_2 are 1.3g/CC and 1.6g/CC.

3. Purity and physical state of the metal

Generally pure metals show a greater resistance to corrosion compared to impure metal. For example- Zn containing about 0.05% of C, Pb or Fe as impurity corrodes several thousand times faster than pure Zn. The impurity facilitates the formation of tiny galvanic cells promoting corrosion. The formation of galvanic cells depends upon the distribution of impurity in the metal. If the distribution is not uniform leading to heterogeneity, then galvanic cells are formed. On the other hand, if the impurity forms homogeneous solid solution (as in alloys) local galvanic cells are not formed.

Physical state here refers to the size of the crystals in the metal. Metals are considered polycrystalline. The size of the crystals (grain size) in them depends upon the conditions of manufacture, specially cooling. Smaller the grain size, more are the grain boundaries and therefore more is the rate of corrosion.

4. pH of the medium: pH is a measure of acidity of the medium. Lesser the pH more acidic is the medium (pH = -log H^+). Metals may be exposed to different H^+ ion concentrations. H^+ ion concentration (lesser the pH), more is the rate of corrosion.

This is due to the following reasons.

(i) H^+ ions help for cathodic reaction (reduction). $2H^+ + 2e^- \longrightarrow H_2$

This causes the anodic reaction (oxidation) to occur. More the H^+ ion concentration more is the rate of cathodic reaction and therefore more is the rate of anodic reaction (corrosion)

(ii) The corrosion products (usually an oxide, hydroxide or carbonate) are basic in nature. They dissolve in acidic environments exposing the metal surface to the environment, for corrosion.

Examples: Metal Zn that corrodes rapidly even in a weakly acidic medium shows good resistance at pH =11.

(iii) Fe reacts rapidly with the liberation of hydrogen below pH=3 (highly acidic). As pH of the medium increases its corrosion rate decreases. Up to pH= 10 it corrodes slowly with the absorption of oxygen. When pH of the medium crosses 10, Fe stops corroding as the insoluble corrosion product [$Fe(OH)_2$] settling on the surface creates a barrier.

5. Temperature of the medium: As temperature increases, corrosion rate also increases for many metals.

i. Increase in conductance of the medium- the species (ions) migrate fast as the temperature increases.

ii. Increase in the solubility of corrosion products- this prevents the formation of a barrier between the metal and the environment.

iii. Increase in the rate of chemical reaction- corrosion involves oxidation and reduction reactions.

iv. Breakdown in the protective film- when temperature increases metal expands more than the barrier layer. This would result in the cracking of the barrier exposing metal for corrosion.

v. Decrease in polarization effects at anodic and cathodic areas.

It is known that solubility of gases (like O_2, CO_2 etc) in a liquid medium decreases with increase in temperature as they show a tendency to escape out. Rate of corrosion due to such gases therefore decreases with increase in temperature.

6. **Relative areas of anode and cathode (Area effect):** According to electrochemical theory the metallic part that acts as anode corrodes. The electrons liberated due to oxidation at anode migrate to the cathodic area where they are consumed through reduction reaction. Whenever a cathodic area is large it can put up a greater demand for electrons. This increases the rate of corrosion, and the rate depends upon how fast the electrons are consumed at cathode.

Therefore, small anodic area and large cathodic area on a metal surface leads to severe corrosion of anodic area (It leads to pitting). On the other hand, if anodic area is large and the cathodic area is small, corrosion still occurs but at a slower rate.

This can be illustrated with the example below (Fig. 3.10).

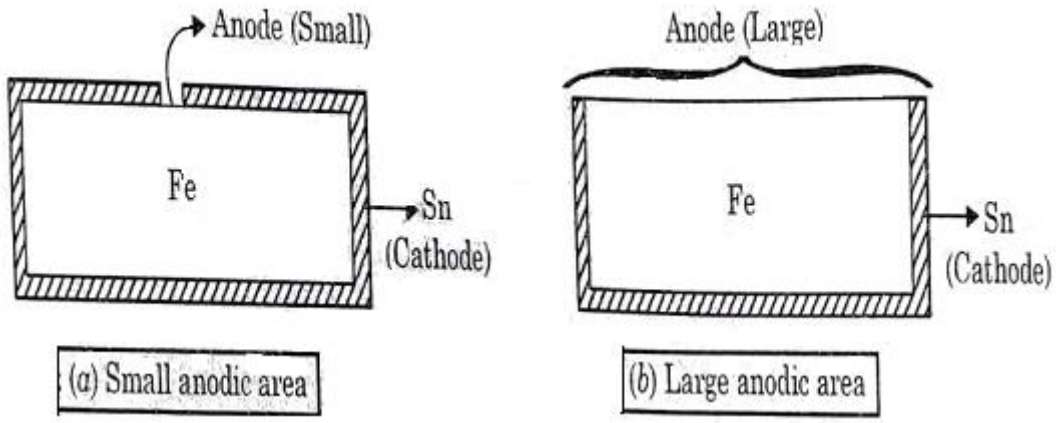


Fig. 3.10: Area effect on corrosion rate

Fe corrodes rapidly in the first case whereas slowly in the second case (observe the relative areas of anode and cathode)

It is important for an engineer to understand this because it gives a clear idea about the selection of metals for designing a structure. His selection should be such that there should be small cathodic areas and large anodic areas in the structure.

CORROSION CONTROL

When use of metals like Au, Pt etc for all purposes is not possible due to limited availability and cost, we have to use metals like Fe, Al, Cu etc. that are available and cheap. But they suffer from the disease called corrosion. The option left is to safeguard them from corrosion by suitable techniques. Some of the important techniques are

1. Coating technique: A metal in contact with a suitable environment corrodes. What happens if the metal is kept out of contact from the environment and how to achieve that? The metal stops corroding if it is kept out of contact, and it can be achieved by creating a barrier between the metal and the environment. Such a barrier is referred to as coating (Fig.3.11). For the coating to be effective in preventing corrosion, it should be **non-porous and adherent. It itself should not react with the environment and should be insoluble.** Further, coatings are chemically different from the metal. Hence, they are helpful in changing the surface properties of the metal like appearance, hardness, electrical conduction or insulation etc.

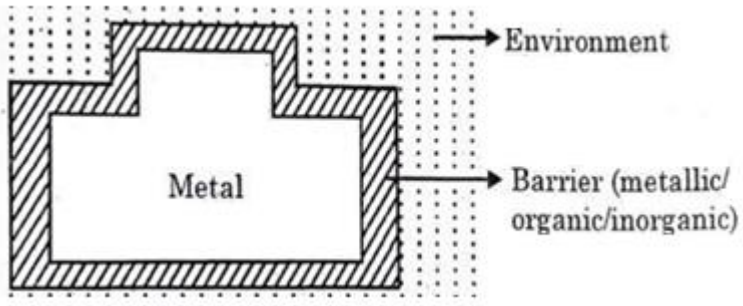


Fig 3.11: Barrier preventing contact between the metal and the environment.

The coating may be **organic, inorganic or metallic** in nature.

Based on this, widely used techniques can be grouped as

(i) Metallic coatings: Examples-galvanization, tinning, chrome plating, gold plating etc.

(ii) Organic coatings: examples-paints, enamels, lacquers

(iii) Inorganic coatings: examples- phosphating, anodizing etc.

Why metallic objects must be cleaned before giving a coating and how they are cleaned?

Any coating (metallic, organic or inorganic) given should adhere firmly to the surface. Then only it can give protection to the base metal for long time. During fabrication, storage and transportation there is every likely hood of some rusting and dust settling. There may be some grease also originated from the manufacturing point (sometimes as a temporary measure,

grease is applied on the metallic objects). If coating is done in presence of these substances, then coating will be non-adherent and so will peel off very easily. So metallic objects must be cleaned before coating. Effective cleaning is achieved through the following processes.

i. Mechanical cleaning – like Grinding, polishing, sand blasting rubbing etc. They are mechanical operations that mainly make the surface even. Some impurities also get removed during these processes.

ii. Removal of oil and grease – As these impurities are organic in nature, they are removed by washing the object with organic solvents like carbon tetrachloride, dichloromethane, chlorobenzene, toluene etc (like dissolves like). Fatty oils can also be removed by washing with NaOH. Fatty oils form soap with NaOH and can be washed away. Mineral oils can be removed by emulsification.

iii. Removal of Rust- rust is basic in nature and therefore reacts with acid. So rust is removed by dipping the object in hot (around 70 ^{0}C) dilute sulphuric acid (5-10%) solution. This process is also called **PICKLING** and it dissolves rust.

iii. Washing with water- finally the object is washed with water and then taken for coating.

1. **Metallic coatings:** It involves coating the base metal with another metal. When two metals are in contact and if there isa chance for corrosion, one of the metal acts as anode and corrodes. Based on how the coating behaves with respect to the base metal, two types of coatings are possible

   Note: Special care has to be taken while giving this type of coatings. The coating should cover the base metal hundred percent. If any part of the base metal is left uncovered or there is a crack in the coating, severe corrosion of the base metal occurs. Small anodic area (base metal) and large cathodic area (coating) further worsens the situation leading to pitting.

(a) Anodic metallic coatings- Here the coating is anodic to the base metal (Fig. 3.12) means that the coating acts as anode with respect to the base metal. The base metal is coated with a more reactive (less noble) metal (higher up in the electrochemical series). When there is a chance for corrosion, the coating acts as anode and corrodes. The base metal obviously acts as cathode and therefore is safe.

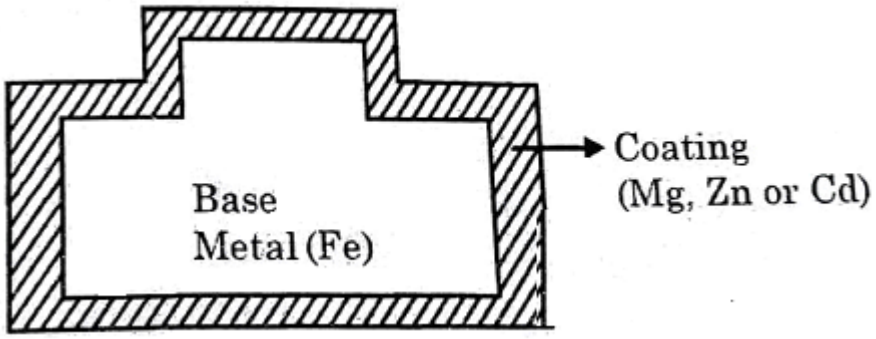


Fig 3.12: Anodic metallic coating

Zinc sheets used for roofing and GI pipes (galvanized iron pipes) used for domestic water supply are the examples. The base metal in both is Fe.

Note: A crack or break in the coating can lead to the formation of galvanic cell. The base metal is still safe as its acts as cathode. The base metal exposed along the cracks is cathode whereas the entire coating is anode. In this situation there will be a large anodic area and a small anodic area and leads to slow corrosion of the coating.

(b) Cathodic metallic coatings: whenever the coating is cathodic to base metal means that it acts as cathode with respect to the base metal then it is called cathodic metallic coating. This is achieved by coating the base metal with another metal lower down (more noble) in the electrochemical series (Figure 3.13)

Example- Coating of Iron with Sn, Ni, Ag, Au etc.

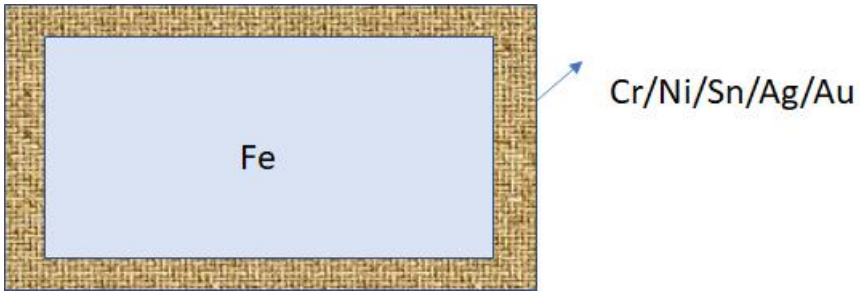


Fig. 3.13: Cathodic metallic coating

Galvanization:

It is an anodic metal coating. It is a process of coating Zn on Fe/steel objects. As zinc has low melting point (419 ^{0}C), galvanization is usually done by hot dipping process.

Hot dipping process –Cleaned object (cleaning steps described earlier) is dipped in molten Zn maintained around 440 ^{0}C. **Fig. 3.14** represents the galvanization process of iron sheet. Cleaned sheet is passed through molten zinc. Molten zinc is kept covered with NH_4Cl flux (**NH_4Cl** flux-prevents

the oxidation of the molten zinc by oxygen of the air***). Excess zinc sticking to the sheet is removed and the thickness of the zinc coating is*** maintained by passing the sheet through rollers with a specific clearance.

Galvanization is done on sheets used for roofing, pipes used for water supply, electric pillars, lamp posts etc.

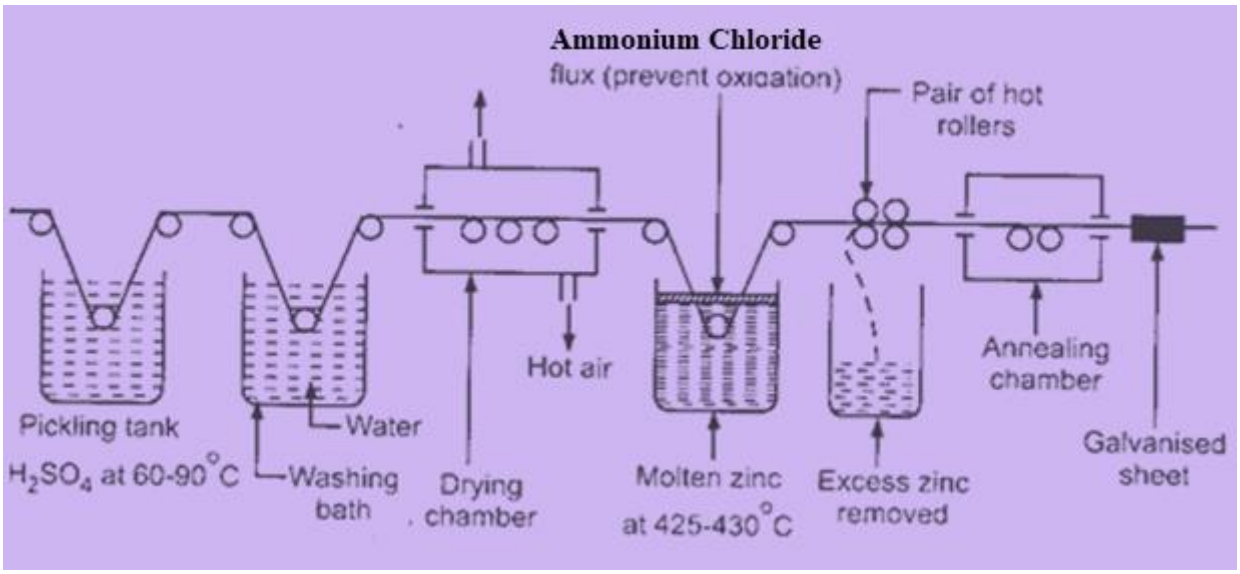


Fig. 3.14: Galvanization of iron sheet

Tinning: it is a process of coating tin on iron or steel. It is a cathodic metallic coating. Hot dipping process is widely used as tin can be melted easily (melting point = 219 ^{0}C)

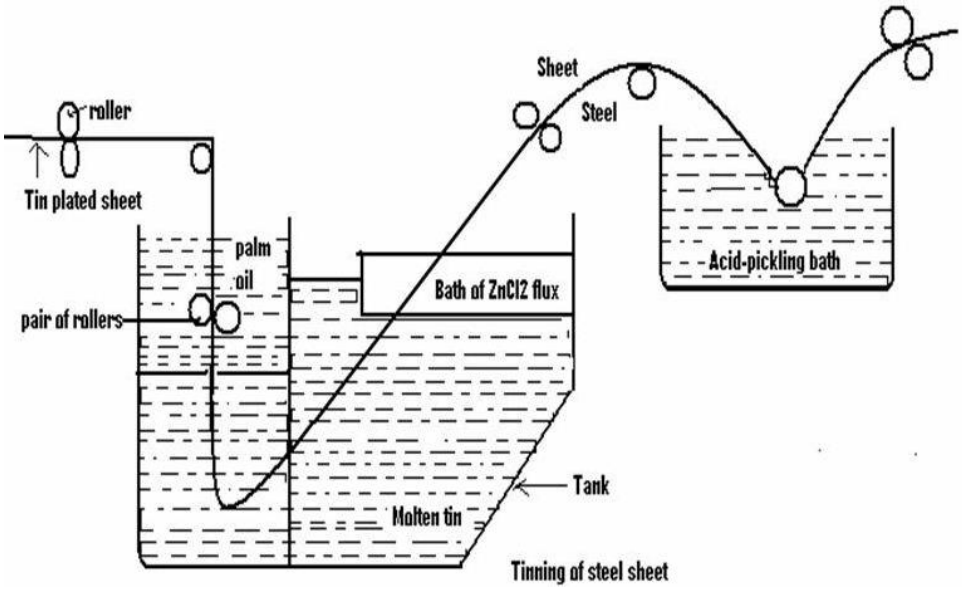


Figure 3.15: Tinning of iron sheets.

Fig.3.15 depicts the hot dipping process of tin on iron sheet. Cleaned iron sheet is passed through

$ZnCl_2$ flux, then through molten tin and finally through palm oil. Rollers are used to maintain the thickness of the coating and to get a uniform coating.

Tinning is done on copper and brass containers used for storing food stuffs and for cooking. Copper wires when tinned become solderable (this has application in PCBs)

Note: Zinc chloride flux helps in the better adherence of coating and palm oil prevents the oxidation of hot tin coating by air.

2. Inorganic coatings: When the barrier coating is made up of an inorganic compound it is called inorganic coating. Such coatings are provided by converting the metal at the surface into an inorganic compound like oxide or phosphate. Hence, they are also called chemical conversion coatings. The coatings are developed by chemical or electrochemical method. Such coatings in addition to providing corrosion resistance have the following additional advantages.

(i) They provide electrical insulating property to the surface and

(ii) They enhance the adherence of paint and enamel to the surface.

Important inorganic coatings are

Phosphate coating or Phosphating -Developing a phosphate layer on the surface by the reaction of the metal with dilute phosphoric acid solution is called phosphating.

Phosphating can be done by

(i) Dipping a cleaned object in phosphating solution

(ii) Spraying phosphating solution in the object or

(iii) Applying the solution using a brush

Constituents of phosphating solution and their function:

(a) Free phosphoric acid- it is the main constituent. It reacts with the metal and produces metal sulphate.

(b) Metal phosphates of zinc and manganese-they help in increasing the adherence of the phosphate layer to the metal surface. They also decrease the porosity of the coating.

(c) Nitrates, chromates, peroxide etc-they are oxidizing agents. Hence, they help in the oxidation of the metal and thus accelerate the formation of phosphate layer/

(d) The pH of the bath is maintained in the range 1.8-3.2.

(e) Temperature of the bath is maintained around 35° C.

The Phosphating involves the following steps:

(i) The dissolution of the metal ie., by undergoing oxidation.

(ii) Metal ions combining with phosphate ions to form metal phosphate.

(iii) Deposition of an insoluble phosphate layer.

$3Fe + 2\ H_3PO_4 \rightarrow Fe_3(PO_4)_2 + 3\ H_2\uparrow$

Properties of phosphate coatings:

i. Coatings are porous and hence cannot guarantee hundred percent protection against corrosion by themselves.

ii. coatings contains zinc and manganese phosphates

iii. Coatings are generally grey in colour but can be given desired colour using dyes.

As phosphate layer is a product of the reaction between metal and phosphoric acid, it binds to the surface well. As it is porous it does not form a perfect coating. But the surface gets the advantage of an adherent painting. When paint is applied directly on a metal surface. It will have a lesser scratch resistance. When it is applied on a phosphated surface, it penetrates into the pores and gets locked. This produces a scratch resistance paint film. Hence it is primarily used as an undercoat to paint or enamel.

Applications:

Phosphate coating is given as an under layer (under coat) before painting the car bodies, refrigerators and washing machines.

Phosphate coatings are generally rough and porous. Hence, they serve as a good undercoat to paint. Phosphating is done on nails, screws, bolt-nuts, drums etc. made out steel.

Anodizing of Aluminium or anodized coatings

Metals like Al undergo passivation by forming an oxide layer. Formation of an oxide layer is an oxidation process, and it occurs at anode. So, the oxide layer can be developed by making the object anode (hence the name anodizing). **Anodizing is a process of developing an oxide layer on a metallic object by induced oxidation, making the object anode and passing current.**

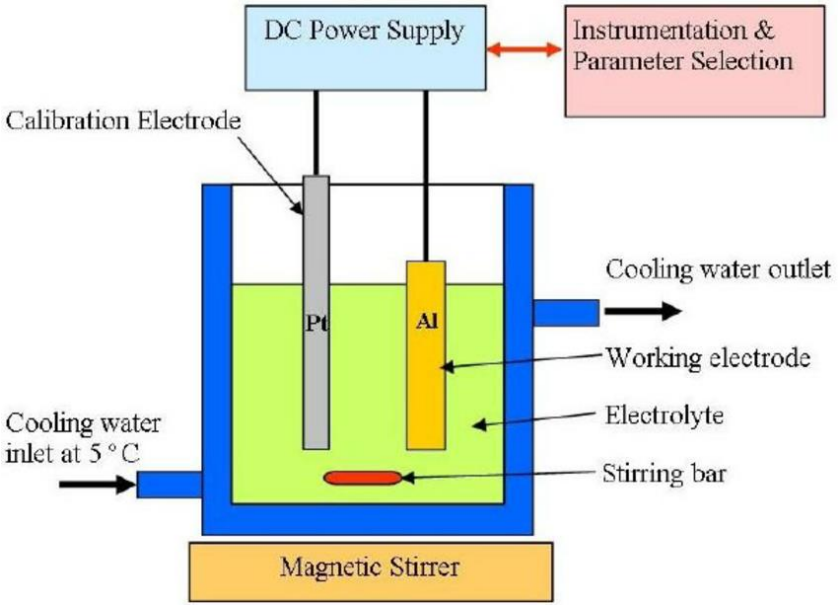


Figure 3.16: Anodization of aluminum using acid electrolytes.

Electrodes:
Anode: Cleaned Aluminum article,
cathode: Lead/copper/steel
Bath conditions:
Electrolyte: 5-10 % chromic acid/ Oxalic acid, Temperature of the bath: 40 °C, voltage-40 V.

The Electrode reactions occurring during anodizing of Al process are:

At anode:

$$2\ Al + 3\ H_2O \longrightarrow Al_2O_3 + 6\ H^+ + 6e^-$$

At cathode:

$$6\ H^+ + 6\ e^- \longrightarrow 3H_2\uparrow$$

Overall reaction:

$$2\ Al + 3H_2O \longrightarrow Al_2O_3 + 3\ H_2\uparrow$$

On anodizing, Al_2O_3 is formed on the surface as a porous layer. The layer may be made compact by **sealing**.

Sealing: The alumina layer on Anodized aluminum will have pores. The pores are blocked by a process known as sealing. It involves dipping the anodized object in hot water (~ 70 ^{0}C) or exposing it to steam. Al_2O_3 undergoes hydration to Al_2O_3. H_2OHydrated alumina occupies higher volume. Due to this there will be expansion of the layer and the pores get covered.

Note: Other metals such as Mg, Ti etc. can also be anodized.

Applications: Metals such as Al, Mg, Ti etc. are anodized to control corrosion.

Cathodic Protection:

In cathodic protection electrons are provided from an external source so that the metal or alloy remains as cathode.

Sacrificial Anode Protection

- In this method the metal to be protected from corrosion is converted into cathode by connecting into a metal which is anodic to it.
- The base metal is usually iron, copper or brass.
- Metals like Mg, Al and Zn are more active and hence are used as anodes.
- These metals being more active acts as anode undergo corrosion and supply electrons to the specimen.
- In this way the protected metals act as cathode.
- Since anodic metals are scarified to protect the metal structure this technique is called sacrificial anode method.
- Example: Mg block connected to buried oil storage tanks,
- Zn wire fixed to the sides of ocean-going ships are the best examples of sacrificial anode methods.

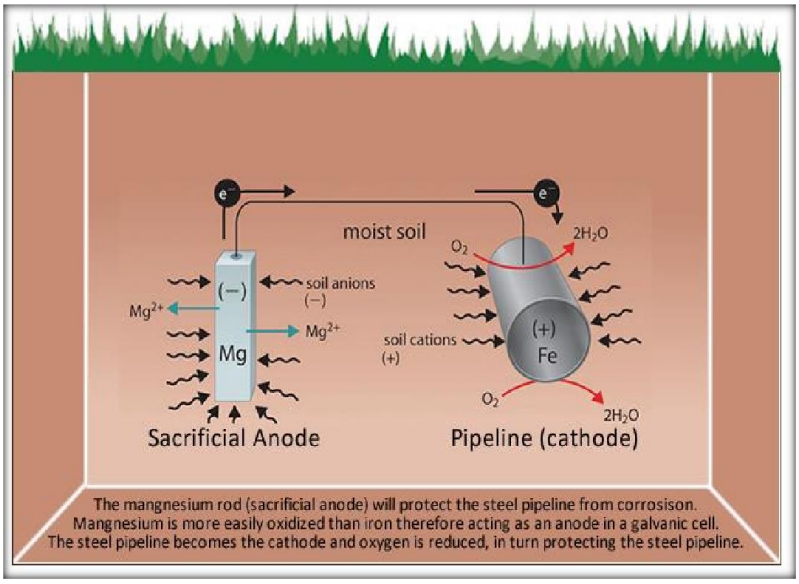


Fig 3.17: Sacrificial anode method

Corrosion testing by Weight-loss method

1. Weight-loss tests are mostly used for measurement of corrosion.

2. In this, clean metal coupon or a standard test piece is measured, weighed, and exposed to the corrodent for a known time.

3. The piece is then taken out, cleaned to remove the corrosion products and reweighed.

The rate of corrosion of the metal, R is then calculated from the equation.

$$\mathbf{CPR = KW / ATD}$$

Where **K** is constant,

T is time of exposure to the nearest 100th of an hour.

W is the weight loss to the nearest milligram

A is the area to the nearest sq.cm, and

D is density, g/cm^3.

CPR is corrosion penetration rate

The result is expressed in many different units such as $mg/dm^2/day$ or $oz/ft^2/day$ or month or inches per year (IPY) or millimeters per year (MPY). Assuming that the surface corrosion was uniform, these units may be converted to depth of corrosion or penetration.

Note: CPR – mpy (mil/per/year) 1mil= 0.001inch hence K=534 should be taken.

CPR- mm/year K= 87.6 should take.

Table 3.1: Corrosion penetration rate values

	CPR in mpy	CPR in mmpy
K	534	87.6
W (wt loss)	mg	mg
(D)	g/cm^3	g/cm^3
A	$Inch^2$	cm^2
t	hrs	hrs

Problem 1: A piece of corroded steel plate was found in a submerged ocean vessel. It was estimated that the original area of the plate was 10 in^2 and that approximately 2.6 kg had corroded away during submersion. Assuming a corrosion penetration rate of 20 mpy for this alloy in sea water, estimate the time of submersion in years. The density of steel is 7.9 g/cm^3.

CPR = KW / AT

A - (area) =10 in2, W- (Weight loss) =2.6 Kg, CPR- (corrosion penetration rate) = 200 mpy

T- (Time taken) =? Years K – (Constant) = 534 KW

After calculation **T= 10.53 years**

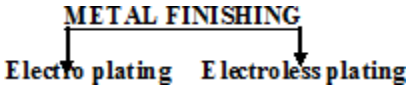
METAL FINISHING

Introduction:

Many metal objects which we see in everyday life like car components, window frames, kitchen utensils, etc., have undergone metal finishing.

Meaning - Metal finishing is the name given to a wide range of processes carried out to modify the surface properties of a metal. The surface properties of a metal may be modified by the deposition of a layer of another metal or a polymer or by formation of an oxide film.

Variety of processes that are performed in order to modify the surface property a metallic object come under metal finishing processes.



It includes painting, galvanization, tinning, anodizing, phosphating, electroplating, electroless plating, electroforming, electro polishing, electrochemical etching Etc.

Importance of metal finishing:

In modern time the importance of metal finishing for purely decorative purposes has decreased. The trend is now towards surface treatments, which will impart:

(i) Corrosion resistance (ii) Particular physical properties to the surface (electrical conductivity, thermal conductivity and magnetic) (iii) Particular mechanical properties to the surface (Hardness, heat and wear resistance and solderability)

(iv) To rebuild the damaged part or worn-out surface of machinery.

(v) In modern engineering -

(a) To manufacture electronic component such PCBs, capacitors and contacts.

(b) To manufacture completely an article or component by electrodeposition Electroforming.

(c) To produce metal article with a highly reflective Mirror finish - Electro polishing.

(d) To manufacture of semiconductors in microelectronics fabrication - Electrochemical etching.

(e) To provide corrosion resistance and decorative look- electroplating

Electroplating process: -

It is a process of deposition of a metal or alloy on a conducting surface through the process of electrolysis.

OR

It is a process of deposition of a metal or alloy on a conducting surface by passing direct current through the solution containing the metal ions to be deposited and making the

object cathode.

The principal components of an electroplating process are depicted in the figure. The essential components include:

1. An electroplating bath containing a conducting salt which furnishes the ions of the metal to be deposited.
2. The electronically conducting anode which may be soluble or insoluble. Generally, the anode consists of the metal to be deposited.
3. The electronically conducting cathode i.e., the object to be plated.
4. An inert vessel containing the electrodes and aqueous electrolyte and is usually a Steel, rubber lined steel, polypropylene or polyvinylchloride vessel.
5. A D.C. electric power source, usually a regulated transformer/rectifier

Example:

In electroplating of copper, the coating metal copper is made anode. During electrolysis, the copper anode dissolves and Cu^{2+} ions enter into the solution. At cathode the Cu^{2+} ions from the solution are discharged and get deposited in the surface of the object to be electroplated. Since the loss of Cu^{2+} ions from the solution are supplied from the anode reaction, the concentration of electrolyte in the bath remains constant.

Reactions are

$$CuSO_4 \longleftrightarrow Cu^{2+} + SO_4^{2-}$$

On passing current Cu^{2+} go to the cathode and gets discharged and deposit on the article.

At anode: $Cu \longrightarrow Cu^{2+} + 2e^-$

At cathode: $Cu^{2+} + 2e^- \longrightarrow Cu$

However, if the anode is made of same inert material that does pass into solution, the electrolyte solution is added continuously in order to maintain optimum metal ion concentration in the bath.

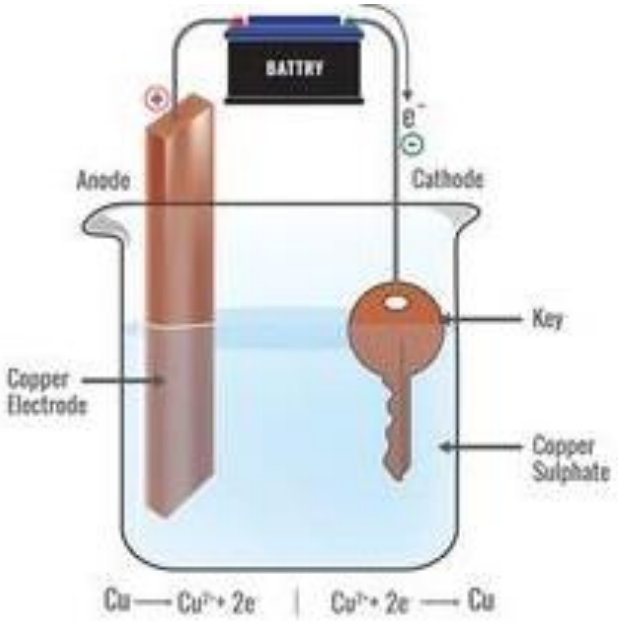


Figure 3.18: Electro plating of copper

Types of plating

Chromium plating /Chrome Plating

Anode : Pb-Sb or Pb- Sn alloy coated with PbO_2

Cathode : Object

Electrolyte : CrO_3 450 g / dm^3 H_2SO_4 4 g / dm^3

Additives : Complex fluorides to increase efficiency Temperature: 45-60 ^{0}C

Current density : 100 - 200 mA / cm^2

Current efficiency : 8-12 %

Hardness : 800-1000 HV

Chromium anodes are not used in chromium plating. This is because, the acidity of chromium plating bath is very high and hence the metal chromium dissolves rapidly resulting in a very high concentration of chromium ions which is undesirable.

The object is given an undercoat of Cu or Ni prior to plating with Cr. If chromium is directly electroplated, the deposits will be porous, and a network of cracking is developed.

Characteristics of chromium coating: -

(i) Hard and wear resistant.

(ii) Corrosion - resistant.

(iii) Low friction and non-stick.

Engineering applications

Chromium plating finds application in the reclamation of worn or over size components, hydraulic rams, rolls printing.

Electroless plating:

Electroless plating is the controlled autocatalytic deposition of a continuous film of a metal from its salt solution by a reducing agent without using electrical energy.

Electroless plating can be used for special applications for which conventional electrode position processes are unsuitable.

Advantages of electroless plating:

1. Deposition on non-conductors, for example, plastics is possible.
2. The baths have excellent throwing power - can be plated inside the tubes, recesses, etc.
3. Excellent corrosion resistance.
4. Electroless coatings are harder than conventional electroplated articles and as a result resistance is increased.
5. No levelling is increased.

6 The deposits have unique chemical, mechanical and magnetic.

Table 3.2: Distinction between electroplating and electroless plating

No.	Property	Electroplating	Electroless plating
1	Driving force	Power supply	Autocatalytic redox reaction
2	Anode reactant	Metal	R, reducing agent in solution
3	Site of anode reaction	Separate anode	Substrate
4	Site of cathode reaction	Substrate	Substrate having a catalytic surface
5	Anode reaction	Oxidation of metal	Oxidation of a soluble reducing agent
6	Cathode reaction	$M^{n+} + n\,e \longrightarrow M$	$M^{n+} + n\,e \longrightarrow M$

Electroless plating of Cu

Modern electroless copper films have a ductility and conductivity identical to that to that of electrolytic copper. The major technique used to plate copper on PCBs (i.e., to manufacture PCBs) is the subtractive method. Printed circuit boards are composed of epoxides, phenolics, flexible films of polyester and polyimide. Copper foil is laminated to both sides of a flat board. Through holes are punched or drilled.

The article (i.e., board) is first treated with organic solvents and alkaline solutions to remove grease and then etched in an acid. It is then activated by a specialized process i.e., dipping first in $SnCl_2$ and then in $PdCl_2$ solutions. It is placed in an electroless copper bath when a thin conductive layer of copper deposits on the entire board including the non-conducting holes.

The plating bath has the following.

$CuSO_4$. $5H_2O$ ($12g/dm^3$), HCHO ($8g/dm^3$), NaOH ($15g/dm^3$), Rochelle salt ($14g/dm^3$),EDTA ($20g/dm^3$); pH = 11; Temperature: 25 ^oC. Reactions:

Ionization :

$CuSO_4 \longrightarrow Cu^{++} + SO_4^{--}$ Oxidative reaction:

$2HCHO + 4OH^- \longrightarrow 2HCOO^- + H_2 + 2H_2O + 2e^-$

Reduction reaction: $Cu^{++} + 2e^- \longrightarrow Cu$

Overall reaction: $2HCHO + 4OH^{--} + Cu^{2+} \longrightarrow 2HCOO^- + Cu + H_2 + 2H_2O$

Applications:

The major application of electroless copper plating is in the manufacture of PCBs. The PCBs find use in communications, instruments, controls, aerospace, etc.